

LiHoF₄ as a spin-half non-standard quantum Ising system

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LiHoF₄ is a magnetic material known for its Ising-type anisotropy, making it a model system for studying quantum magnetism. However, the theoretical description of LiHoF₄ using the quantum Ising model has shown discrepancies in its phase diagram, particularly in the regime dominated by thermal fluctuations. In this study, we investigate the role of off-diagonal dipolar terms in LiHoF₄, previously neglected, in determining its properties. We analytically derive the low-energy effective Hamiltonian of LiHoF₄, including the off-diagonal dipolar terms perturbatively, both in the absence and presence of a transverse field. Our results encompass the full $B_x - T$ phase diagram, confirming the significance of the off-diagonal dipolar terms in reducing the zero-field critical temperature and determining the critical temperature's dependence on the transverse field. We also highlight the sensitivity of this mechanism to the crystal structure by comparing our calculations with the Fe₈ system.

Introduction.—LiHoF₄, a magnetic material with remarkable properties, has become a focal point for research seeking to unravel the intricacies of quantum magnetism. Known for its distinct Ising-type anisotropy, LiHoF₄ serves as an archetype among anisotropic dipolar magnets, encompassing single-molecule magnets and rare-earth magnetic insulators. These magnets, characterized by a substantial anisotropy barrier and a ground state doublet, provide an intriguing test bed for studying perhaps the simplest model for (quantum) magnetism—the (transverse field) Ising model [1–3]. In addition to their fundamental significance, anisotropic dipolar magnets have garnered attention due to their potential applications in nanomagnets, qubits, and memory bits [4–6]. In these regards, LiHoF₄ has emerged as one of the most extensively studied materials in this category, providing insights into diverse phenomena such as quantum tunneling [7], quantum criticality [1, 8–10], quantum annealing [11], the effects of disorder [12–17], quantum entanglement [18], the formation and dynamics of magnetic domains [19–21], and high-Q nonlinear dynamics [22]. However, despite being regarded as a quintessential Ising magnet, the persistent discrepancy in its $B_x - T$ phase diagram compared to the predictions of the transverse-field Ising model (TFIM), particularly in the high-temperature, low-field regime dominated by thermal fluctuations, calls into question the adequacy of the theoretical description of LiHoF₄ in terms of the TFIM.

Dipolar interactions dominate LiHoF₄, and over the years, there has been increasing recognition of the role of the off-diagonal terms of the dipolar interaction in determining the properties of the material, particularly its diluted series, LiHo_xY_{1-x}F₄, where holmium ions are randomly replaced by non-magnetic yttrium ions. In LiHo_xY_{1-x}F₄, the interplay of an applied transverse field, random disorder, and the off-diagonal dipolar (ODD) terms has been shown to reduce the symmetry of the system, rendering the TFIM an inadequate model for the system [23–27]. Nevertheless, when studying the pure LiHoF₄ system, these ODD terms were thus far usually neglected in deriving an effective low-energy Hamil-

tonian, citing their smallness and vanishing mean-field (MF) contribution [28, 29].

In [30], we suggested that the ODD terms are of quantitative importance in determining the phase diagram, and numerically showed that their inclusion in the model results in a notable reduction of the low-field critical temperature. Here we analytically derive the effective low-energy Hamiltonian of the pure LiHoF₄ system, perturbatively including the off-diagonal dipolar terms, both in the absence and in the presence of a transverse field. Thus, we find that ODD terms manifest in the effective low-energy model as three-body interactions with an overall anti-ferromagnetic preference. Our results confirm the role of the off-diagonal terms in reducing the critical temperature, and the dependence of the mechanism on the transverse field, thus enabling the derivation of the full phase diagram of LiHoF₄ in transverse field. Intriguingly, this mechanism of reduction of the critical temperature due to the contribution of the off-diagonal dipolar terms sensitively depends on the structure of the crystal. To emphasize this point, we repeat our calculations for the Fe₈ system and show that in this system, the reduction of T_c is negligible, in agreement with experiment.

Theory.—The dominant interactions in LiHoF₄ are magnetic dipolar interactions between the Ho³⁺ ions. Combined with its crystal structure, these result in ferromagnetic order below $T_c = 1.53$ K. Its strong single-ion easy-axis anisotropy results in an Ising-like twofold degenerate non-Kramers ground state and a first excited state separated by an energy gap $\Delta \approx 11.5$ K. Previous works aiming to derive an effective low-energy Hamiltonian did so by directly projecting the full microscopic Hamiltonian onto the low-energy subspace spanned by its two lowest-energy states (the ground state doublet is continuously split by the transverse field B_x) [28, 29]. In the absence of an applied transverse field, this approach immediately reduces the dipolar interaction to a strictly longitudinal interaction since the x and y electronic angular momentum operators have vanishing matrix elements between the Ising-like ground states. Only at non-zero transverse fields does this projection include

terms proportional to off-diagonal dipolar elements, but in this case, they were neglected due to their smallness compared to the longitudinal interaction and the direct interaction with the transverse field. Thus, this projection results in a simple TFIM Hamiltonian without any contribution from ODD terms.

Here we derive an effective low-energy spin- $\frac{1}{2}$ Hamiltonian that perturbatively includes off-diagonal dipolar terms by applying a Schrieffer-Wolff transformation to the full LiHoF₄ Hamiltonian prior to the projection, treating the interaction terms as a perturbation. This approach was used by Chin and Eastham [31] but only for $B_x = 0$ and considering only the lowest excited state. Here we generalize their approach for arbitrary transverse field and include all excited crystal-field states. We also introduce an exchange interaction between nearest neighbors of strength J_{ex} . The Hamiltonian is divided into an unperturbed diagonal part, H_0 , and a perturbation, H_T , which consists of the dipolar and exchange interaction terms, i.e., $H_{\text{full}} = H_0 + H_T$, where

$$H_0 = \sum_i V_c + g_L \mu_B B_x \sum_i J_i^x, \quad (1)$$

$$H_T = \frac{1}{2} E_D \sum_{\substack{i \neq j \\ \nu, \mu}} V_{ij}^{\nu\mu} J_i^\nu J_j^\mu + J_{\text{ex}} \sum_{\langle i, j \rangle} J_i^\nu J_j^\nu \quad (2)$$

with $\mu, \nu \in \{x, y, z\}$. The hyperfine interaction with the nuclear spins is treated separately, as described later. The dipolar interaction $V_{ij}^{\mu\nu} = [\delta^{\mu\nu} r_{ij}^2 - 3(\vec{r}_{ij})^\mu (\vec{r}_{ij})^\nu] / r_{ij}^5$, and $k_B E_D = \frac{\mu_0 \mu_B^2 g_L^2}{4\pi}$. The crystal field parameters that comprise V_c are those suggested in Ref. [32], which are identical to Ref. [33] except for a small adjustment of $B_6^4(s)$. The low-energy subspace onto which H_{full} is projected is that in which all of the ions are in either of their two lowest-energy eigenstates induced by H_0 ; the appropriate projection operator is denoted P_0 . The Schrieffer-Wolff transformation uses a generator S , in terms of which the projected low-energy Hamiltonian is given as [34]

$$H_{\text{eff}} = H_0 P_0 + P_0 H_T P_0 + \frac{1}{2} P_0 [S, H_T] P_0 + \mathcal{O}(H_T^3) \quad (3)$$

Details of the calculation are found in the supplemental material [35].

Three-state model in zero-field.—For concreteness, we perform the procedure explicitly for $B_x = 0$ while considering only the first excited state. This means that before the Schrieffer-Wolff transformation is applied, all operators of the full Hamiltonian are truncated so that they only act upon the three lowest-energy states. The two lowest-energy states, which in the absence of a transverse field are designated $|\uparrow\rangle$ and $|\downarrow\rangle$, are chosen such that $\langle \uparrow | J^z | \downarrow \rangle = \langle \downarrow | J^z | \uparrow \rangle = 0$. The operators are then given

in matrix form by

$$V_c = \begin{pmatrix} 0 & & \\ & 0 & \\ & & \Delta \end{pmatrix}; \quad J^z = \begin{pmatrix} -\alpha & & \\ & \alpha & \\ & & 0 \end{pmatrix};$$

$$J^x = \begin{pmatrix} 0 & 0 & \rho \\ 0 & 0 & \rho \\ \rho & \rho & 0 \end{pmatrix}; \quad J^y = \begin{pmatrix} 0 & 0 & -i\rho \\ 0 & 0 & i\rho \\ i\rho & -i\rho & 0 \end{pmatrix} \quad (4)$$

where $\Delta = 11.5$ K, and eigenbasis of V_c is chosen such that $\rho = 2.34$ and $\alpha = 5.53$ are real. With these simplifications, we are essentially reproducing the results of Ref. [31], and indeed we obtain an identical result (assuming $J_{\text{ex}} = 0$),

$$H_{\text{eff}}(B_x = 0) = \frac{\alpha^2}{2} E_D \sum_{i \neq j} V_{ij}^{zz} \sigma_i^z \sigma_j^z + \alpha^2 J_{\text{ex}} \sum_{\langle i, j \rangle} \sigma_i^z \sigma_j^z$$

$$+ \sum_i h_i^\mu \sigma_i^\mu + \sum_{\substack{i \neq j \\ \nu, \mu}} \varepsilon_{ij}^{\mu\nu} \sigma_i^\mu \sigma_j^\nu + H_{3B} \quad (5)$$

where

$$H_{3B} = -\frac{\alpha^2 \rho^2}{\Delta} E_D^2 \sum_{i \neq j \neq k} (V_{ij}^{xz} V_{ik}^{xz} + V_{ij}^{yz} V_{ik}^{yz}) \sigma_j^z \sigma_k^z$$

$$- \frac{\alpha^2 \rho^2}{\Delta} E_D^2 \sum_{i \neq j \neq k} (V_{ij}^{xz} V_{ik}^{xz} - V_{ij}^{yz} V_{ik}^{yz}) \sigma_i^x \sigma_j^z \sigma_k^z$$

$$- \frac{\alpha^2 \rho^2}{\Delta} E_D^2 \sum_{i \neq j \neq k} (V_{ij}^{xz} V_{ik}^{yz} + V_{ij}^{yz} V_{ik}^{xz}) \sigma_i^y \sigma_j^z \sigma_k^z \quad (6)$$

gathers three-body terms. σ_i^μ are Pauli matrices acting within the two-dimensional low-energy subspace of each ion i . The full forms of h_i^μ and $\varepsilon_{ij}^{\mu\nu}$ are found in the supplemental material [35]; the fields h_i^μ all vanish due to symmetry in the undiluted case, and the emergent two-body terms $\varepsilon_{ij}^{\mu\nu}$ are two orders of magnitude smaller than the exchange interaction, which itself is significantly smaller than the dominant longitudinal dipolar interaction. The emergent two-body terms are also short-range interactions by virtue of being proportional to the squared dipolar interaction, and thus $\sim 1/r^6$. In contrast, the three-body terms are of the same order of magnitude as the exchange interaction, and, crucially, they are effectively long-range interactions despite being proportional to a product of dipolar interactions due to the additional summation. Thus, the three-body terms are expected to be the most impactful of the new emergent interactions.

Next, we employ a mean-field approximation by neglecting correlation terms $(\sigma_i^\mu - \langle \sigma_i^\mu \rangle)(\sigma_j^\nu - \langle \sigma_j^\nu \rangle) \approx 0$ for any μ, ν and $i \neq j$, and denoting $\langle \sigma_j^\mu \rangle \equiv m_\mu$. In general, we then have three coupled self-consistency equations for $\vec{m}$, but, in the case of zero transverse field, the last two terms in (6) vanish in mean-field due to

lattice symmetries, leaving us with the usual Ising self-consistency equation $m_z = \tanh(\beta b_i^z)$. The exact form of $\vec{b}(\vec{m})$ is given in the supplemental material [35]. The critical temperature determined by this equation, with $q = 4$ as the number of nearest neighbors, is

$$T_c(B_x = 0) = -\alpha^2 E_D \sum_{j(\neq i)} V_{ij}^{zz} - \alpha^2 q J_{\text{ex}} - \frac{\rho^4 J_{\text{ex}}}{\Delta} \left(2E_D \sum_{j \in NN(i)} V_{ij}^{xx} + q J_{\text{ex}} \right) - \frac{\rho^4}{\Delta} E_D^2 \sum_{j(\neq i)} \left[V_{ij}^{xx} V_{ij}^{yy} - (V_{ij}^{xy})^2 \right] + \frac{4\alpha^2 \rho^2}{\Delta} E_D^2 \sum_{k \neq j(\neq i)} V_{ij}^{xz} V_{jk}^{xz} \quad (7)$$

where the notation $j \in NN(i)$ indicates that j sums over the q nearest neighbors of spin i . The last sum in Eq. (7) above, the result of the three-body interaction (6), is the predominant correction to the longitudinal interaction that results in a reduction of T_c ; the two preceding terms are much smaller in value. Fig. 1(a) shows a group of three spins strongly affected by the three-body interaction in such a manner that generates an effective anti-ferromagnetic interaction between two of them. Fig. 1(b) illustrates the spectrum of the three states used in the three-state model.

The (long-range) sums in Eq. (7) are calculated using Ewald's method [48, 49], assuming a needle-shaped domain that, in the non-zero transverse field case to be discussed later, is embedded within a spherical sample whose magnetization in the x direction in response to the transverse field is homogeneous [28]. See further discussion in the supplemental material [35]. Thus we find that for the three-state model, the three-body term is responsible for a 5% reduction of the MF critical temperature. Subsequent sections will demonstrate, by incorporating higher excited crystal-field states and employing Monte Carlo simulations, that this three-body term is actually responsible for a majority of the existing discrepancy between MF predictions and experimental observations.

17-state model in $B_x \geq 0$.—As mentioned previously, the Schrieffer-Wolff procedure and subsequent projection can be performed for arbitrary non-zero B_x , albeit resulting in an excessively complicated effective Hamiltonian compared to (5). Nevertheless, a mean-field approximation can be applied to it just the same, only in this case, the three self-consistency equations are fully coupled, so, to find T_c , they are solved numerically for decreasing T until a non-zero value for $\langle J^z \rangle$ is detected. Here we use the full, rather than truncated, forms of the operators in H_{full} , again using the Schrieffer-Wolff transformation to decouple (up to second order in H_T) the two lowest-energy states from the 15 excited states.

The results on the full phase diagram are shown in Fig. 2; once with the Schrieffer-Wolff transformation ap-

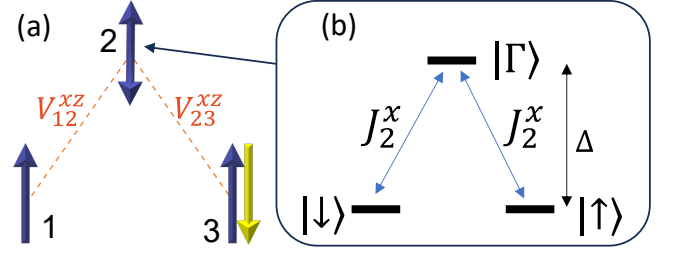


FIG. 1. (a) An illustration of three spins, between two of which (denoted 1 and 3) an effective anti-ferromagnetic interaction emerges due to their off-diagonal dipolar interactions with the third (denoted 2). The prominent three-body term in H_{3B} energetically favors a configuration where spin 3 is anti-aligned with spin 1 (yellow), regardless of the state of spin 2 (thus the double-sided arrow), compared to a configuration where spin 3 is aligned with spin 1 (blue). (b) The three lowest-energy levels of V_c , used in the three-state model. The existence of an excited level $|\Gamma\rangle$ at spin 2, coupled to the ground states $|\uparrow\rangle$ and $|\downarrow\rangle$ by J_2^x , is crucial to the emergence of the three-body interaction, as it modifies the ground state energies of spin 2 in a way that depends on the relative configurations of its neighbors.

plied to the full Hamiltonian in (1) and (2) and once with the Schrieffer-Wolff transformation applied to that Hamiltonian but only with $\mu = \nu$. The results presented include two important modifications. First, following Refs. [32, 50, 51], we rescale the longitudinal interaction by 0.805, i.e., we replace $J_i^z J_j^z \rightarrow 0.805 J_i^z J_j^z$ in the original Hamiltonian prior to deriving the effective low-energy Hamiltonian. This is done to account for the strong c -axis fluctuations not captured in MF [33]. The factor is chosen slightly larger than suggested in Refs. [32, 50, 51] to match the Monte Carlo results of the current work. Of course, a proper account of fluctuations is warranted where possible and is done by Monte Carlo simulation, as described in the next section. Second, following Ref. [28], we include the effect of the hyperfine interaction with the nuclear spins by assuming its predominant effect is a temperature-dependent renormalization of the transverse magnetic field. Even though the hyperfine interaction has since been studied more comprehensively [8, 23, 52, 53], we opted for this simplified approach since the effect of the hyperfine coupling has not been shown to be significant to the region of the phase diagram of highest interest to this work—near the classical transition.

Monte Carlo simulations.—To properly account for fluctuations we perform classical Monte Carlo (MC) simulations on the effective spin- $\frac{1}{2}$ Hamiltonian obtained from the 17-state model at $B_x = 0$, excluding any quantum terms. Since the quantum terms—i.e., terms involving σ_i^x or σ_i^y —are found to have vanishing mean-field contributions, this simulated Hamiltonian, denoted H_{MC} , is effectively equivalent to that used to generate the MF results in Fig. 2. An explicit form for H_{MC} is given in the supplemental material [35]. Naturally, H_{MC} does not

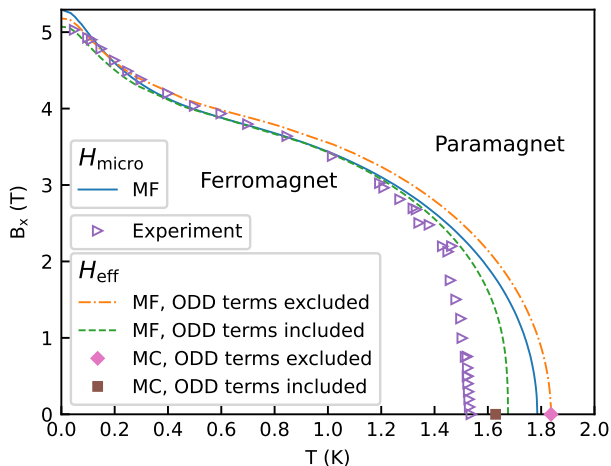


FIG. 2. Phase diagram of LiHoF₄. The solid blue line is the MF result of the full microscopic Hamiltonian with the mean-field scaled by 0.785 from Ref. [51], triangles show experimental results [1, 51, 54], and all other results are based on the effective Hamiltonian H_{eff} derived in this work by a Schrieffer-Wolff transformation from the 17-state model. The dot-dashed line represents the MF phase diagram of H_{eff} derived with the ODD terms explicitly *excluded*, while the dashed line represents MF results with the ODD terms *included*. In both, the longitudinal interaction is scaled by 0.805, as explained in the text. These MF phase diagrams derived from the spin- $\frac{1}{2}$ effective Hamiltonian are corrected for the renormalization of the transverse magnetic field due to the hyperfine interaction, as described in the text. Additionally, results from Monte Carlo simulations at zero-field are represented by a square and a diamond for when ODD terms are included and excluded, respectively. The difference between the two—the result of the three-body term—is in excellent agreement with that found using the more involved MC simulations of Ref. [30].

require any renormalization of the interactions, and, interestingly, the critical temperature of H_{MC} is found to be slightly smaller than its MF critical temperature, even when the MF includes the scaling factor purported to account for fluctuations—see Fig. 2.

We stress that unlike the simulation in Ref. [30], which involved diagonalization of the full single Ho³⁺ ion, the current simulation is a simple Metropolis MC that works on the effective spin- $\frac{1}{2}$ Hamiltonian H_{MC} . In comparing the $B_x = 0$ results of the two, we observe good agreement, attributing the existing deviations to the adjustment in the crystal-field parameters and the much larger system sizes achieved in this study. This agreement indicates that the quantum fluctuations that dictate the reduction of T_c are very well captured by a *classical* three-body term, namely the first term in $H_{3\text{B}}$ (6).

Fe₈ molecular magnet.—Another material considered a good physical magnetic realization of the transverse field Ising model is the molecular magnet Fe₈ in crystal form [3, 55–59]. Interestingly, it does not show an ob-

vious discrepancy between experimental results and MF calculations [3], as does LiHoF₄, raising questions regarding the universality of the effect henceforth described. To address these questions, in this section, we apply the same treatment to Fe₈ as we did for LiHoF₄.

The Fe₈ crystal consists of large molecular clusters, each described by a simple $S = 10$ spin model subject to a strong uniaxial anisotropy described by the crystal field potential $V_{\text{Fe8}} = -DS_z^2 + E(S_x^2 - S_y^2)$ where $D > E$ [55], situated on a triclinic lattice. Experimentally, in the absence of a transverse magnetic field, Fe₈ orders ferromagnetically below a critical temperature of $T_c \approx 0.6$ K [3], in reasonable agreement with theoretical predictions [57, 58].

The only non-negligible interaction between the Fe₈ molecular clusters is dipolar [58], so in the absence of an external transverse field, the full Hamiltonian is given by $\mathcal{H}_{\text{Fe8}} = \sum_i V_{\text{Fe8}} + \frac{\mu_0 \mu_B^2 g^2}{8\pi} \sum_{i \neq j} V_{ij}^{\nu\mu} J_i^\nu J_j^\mu$ with $g = 2$ [55]. When we apply the Schrieffer-Wolff transformation and project $\mathcal{H}_{\text{Fe8}}$ onto the subspace in which all molecules are within their ground state doublet, we obtain an effective low-energy spin- $\frac{1}{2}$ Hamiltonian. This Hamiltonian, similarly to LiHoF₄, consists of longitudinal dipolar interactions, proportional to V_{ij}^{zz} , as well as other two-body and three-body terms, proportional to products of other dipolar components. As before, the emergent three-body interactions dictate a reduction of the MF critical temperature; only now, we find that the contribution of all emergent interactions amount to a negligible correction of less than 0.01%—well below experimental error bars. This is perfectly consistent with the stated lack of discrepancy between MF and experiment. Further details of the calculation are provided in the supplemental material [35].

Discussion.—The discrepancy between theory and experiment regarding the shape of the phase boundary near the classical transition has persisted as an open question over the past two decades [2, 28, 29, 33, 54]. In Ref. [30], it has been shown that the inclusion of ODD terms results in a significant reduction of the critical temperature of LiHoF₄ at low transverse fields. A simplified argument was given for why one could expect this reduction to be attenuated with an increasing transverse field, possibly leading to the experimentally established steep rise of the phase boundary at small fields, thus offering a potential resolution of the discrepancy. Unfortunately, due to the numerical nature of that work, the range of transverse fields for which an exact numerical value could be given was not large enough to clearly show an attenuation of the described mechanism.

In this work, we pursued an analytical approach that allowed us to describe the full $B_x - T$ phase diagram of LiHoF₄ in MF and thus show that the reduction in T_c is indeed attenuated as B_x is increased. As a natural result, the shape of the phase boundary curve obtained from the effective Hamiltonian that includes ODD terms is closer to the experimental curve, showing a similarly steep rise

near the classical transition. We also find that the effect of ODD terms becomes minimal at $B_x \gtrsim 3.5$ T—about the same value where the standard (rescaled) MF and experimental curves merge [51]. This work also provides an explicit analytical expression describing said reduction at $B_x = 0$ (the last term in Eq. (7)), showing that it results from emergent three-body interactions due to ODD terms.

The T_c -reducing effect of ODD terms is not fully captured in our MF analysis, as evidenced by the difference between the $B_x = 0$ MC and MF results when ODD terms are included, as seen in Fig. 2. This difference in critical temperature arises from the mirror symmetry of the crystal, guaranteeing certain cancellations in the homogeneous system assumed in MF, but not in a MC simulation [35]. Notably, the fact that a simple Ising MC simulation agrees reasonably well with the experimental zero-field critical temperature without requiring further special modifications is highly encouraging. Furthermore, the current study represents an improvement over the findings of Ref. [30] as it explicitly acknowledges the omission of quantum terms from the effective Hamiltonian which could account for the remaining discrepancy between the MC simulation and experimental results at $B_x = 0$. Although extending the simulations to $B_x > 0$ would have been desirable, the complexity of the effective Hamiltonian, which in that case contains dozens of non-commuting terms of equal order of magnitude, made any approach beyond MF impractical. Additionally, it should be noted that a further reduction in T_c could be

expected due to the fluctuating transverse spin components not considered in this work; the magnitude of this reduction is yet undetermined [60].

Another interesting feature is the dependence of the mechanism on the crystal structure, made evident by the application of the analysis to the Fe_8 system. Qualitatively, the adjustment of T_c due to ODD terms generally applies to any magnetic anisotropic dipolar system. However, we find that the effect can be quite large, as seen in LiHoF_4 , but may very well be quantitatively negligible, as seen in Fe_8 ; the determining factor being the crystal structure as manifested in MF in sums of the form $\sum_{j \neq k(\neq i)} V_{ij}^{xz} V_{jk}^{xz}$ and $\sum_{j \neq k(\neq i)} V_{ij}^{yz} V_{jk}^{yz}$. In a way, this is not surprising since, as was shown generally by Luttinger and Tisza [61] and later extended specifically to lithium rare-earth tetrafluorides [62], the nature of the ordered phase of a system of dipoles coupled primarily by dipolar interactions is highly sensitive to the geometric arrangement of said dipoles. Thus, the magnitude of the effect of off-diagonal dipolar interactions could vary markedly depending on crystal structure for the same reasons. Perhaps for some lattice configurations, it can also be of opposite sign, i.e., favoring rather than disfavoring ferromagnetic order.

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Supplemental Material: LiHoF_4 as a spin-half non-standard quantum Ising system

I. DERIVATION OF AN EFFECTIVE LOW-ENERGY HAMILTONIAN

As discussed in the main text, the Hamiltonian is divided into an unperturbed diagonal part, H_0 , and a perturbation, H_T , which consists of the dipolar and exchange interaction terms, i.e., $H_{\text{full}} = H_0 + H_T$, where

$$H_0 = \sum_i V_c + g_L \mu_B B_x \sum_i J_i^x, \quad (\text{S.1})$$

$$H_T = \frac{1}{2} E_D \sum_{\substack{i \neq j \\ \nu, \mu}} V_{ij}^{\nu\mu} J_i^\nu J_j^\mu + J_{\text{ex}} \sum_{\langle i, j \rangle} J_i^\nu J_j^\nu \quad (\text{S.2})$$

with $\mu, \nu \in \{x, y, z\}$, the dipolar interaction $V_{ij}^{\mu\nu}$ given by

$$V_{ij}^{\mu\nu} = \frac{\delta^{\mu\nu} |\vec{r}_i - \vec{r}_j|^2 - 3 (r_i^\mu - r_j^\mu) (r_i^\nu - r_j^\nu)}{|\vec{r}_i - \vec{r}_j|^5}, \quad (\text{S.3})$$

and $E_D = \frac{\mu_0 \mu_B^2 g_L^2}{4\pi k_B}$. We designate the two lowest-energy eigenstates of the single-ion H_0 as $|\alpha\rangle$ and $|\beta\rangle$ and define the many-body low-energy subspace through the projection operator $P_0 = \prod_i (|\alpha_i\rangle \langle \alpha_i| + |\beta_i\rangle \langle \beta_i|)$, i.e., it is the subspace in which all of the ions are in one of their two respective low-energy single-ion electronic states. Since H_T consists of two-body operators, it couples the low-energy subspace to either the subspace in which there is a single ion in its excited state or to that in which there are two.

The low-energy Hamiltonian is then given by [34]

$$H_{\text{eff}} = H_0 P_0 + P_0 H_T P_0 + \frac{1}{2} P_0 [S, (H_T)_{od}] P_0 + \mathcal{O}(H_T^3) \quad (\text{S.4})$$

where S is called the generator of the transformation and is given by

$$S = \sum_{i,j} \frac{\langle i | (H_T)_{od} | j \rangle}{E_i - E_j} |i\rangle \langle j|.$$

The notation $(\cdot)_{od}$ indicates that only the block-off-diagonal parts of the operator are considered, that is, $(V)_{od} = P_0 V (\mathbb{I} - P_0) + (\mathbb{I} - P_0) V P_0$.

For the given H_T , we can write S explicitly using projection operators,

$$S = \frac{E_D}{2} \sum_{i \neq j} \sum_{\mu, \nu} \sum'_{\alpha_1, \alpha_2, \beta_1, \beta_2} \frac{V_{ij}^{\nu\mu} P_i^{\alpha_1} J_i^\mu P_i^{\beta_1} \otimes P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}}{E_{\alpha_1} - E_{\beta_1} + E_{\alpha_2} - E_{\beta_2}} \\ + \frac{J_{ex}}{2} \sum_{i \neq j} \sum_{\nu} \sum'_{\alpha_1, \alpha_2, \beta_1, \beta_2} \delta_{ij, nn} \frac{P_i^{\alpha_1} J_i^\nu P_i^{\beta_1} \otimes P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}}{E_{\alpha_1} - E_{\beta_1} + E_{\alpha_2} - E_{\beta_2}}$$

where the single-ion projection operators are defined by

$$P_i^\alpha \equiv \prod_{j(\neq i)} \mathbb{1}_j |\alpha_i\rangle \langle \alpha_i|$$

It should be noted that $\sum_\alpha P_i^\alpha = \mathbb{I}$ is the identity operator of the complete system. Further, we denote E_α as the single-ion eigenenergy corresponding to the single-ion eigenstate $|\alpha\rangle$. The prime on the sums above indicates that we exclude terms for which $\alpha_1, \beta_1, \alpha_2, \beta_2 \leq 2$ or $\alpha_1, \beta_1, \alpha_2, \beta_2 > 2$ or $\alpha_1, \beta_2 > 2 \wedge \alpha_2, \beta_1 \leq 2$ or $\alpha_1, \beta_2 \leq 2 \wedge \alpha_2, \beta_1 > 2$ or $\alpha_1, \beta_1 \leq 2 \wedge \alpha_2, \beta_2 > 2$ or $\alpha_2, \beta_2 \leq 2 \wedge \alpha_1, \beta_1 > 2$. Thus, only block-off-diagonal elements of H_T , i.e., elements that couple states with a different number of spins outside of the low-energy subspace, are included.

The first two terms in (S.4) are the naive projections of the full Hamiltonian onto the low-energy subspace and are trivially calculated. Therefore, we focus on the third term. In the interest of clarity, we split the expression for $\frac{1}{2} [S, H_T]$ into four parts; one for each of the combinations of dipolar and exchange interactions,

$$\frac{I}{E_D^2} = \sum_{k \neq l} \sum_{i \neq j} \left[V_{ij}^{\nu\mu} P_i^{\alpha_1} J_i^\mu P_i^{\beta_1} P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, V_{kl}^{\sigma\tau} J_k^\sigma J_l^\tau \right] \\ \frac{II}{E_D J_{ex}} = \frac{1}{3} \sum_{k \neq l} \sum_{i \neq j} \left[V_{ij}^{\nu\mu} P_i^{\alpha_1} J_i^\mu P_i^{\beta_1} P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, \delta_{kl, nn} J_k^\sigma J_l^\tau \right] \\ \frac{III}{E_D J_{ex}} = \frac{1}{3} \sum_{k \neq l} \sum_{i \neq j} \left[\delta_{ij, nn} P_i^{\alpha_1} J_i^\nu P_i^{\beta_1} P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, V_{kl}^{\sigma\tau} J_k^\sigma J_l^\tau \right] \\ \frac{IV}{J_{ex}^2} = \frac{1}{9} \sum_{k \neq l} \sum_{i \neq j} \left[\delta_{ij, nn} P_i^{\alpha_1} J_i^\nu P_i^{\beta_1} P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, \delta_{kl, nn} J_k^\sigma J_l^\sigma \right]$$

where $\delta_{ij, nn}$ is zero except when spin i and j are nearest neighbors. The factor of $1/3$ takes care of the extra sum over spatial coordinates below that is only relevant to the dipolar terms. Thus,

$$\frac{1}{2} [S, H_T] = \frac{1}{8} \sum_{\mu, \nu, \sigma, \tau} \sum'_{\alpha_1, \alpha_2, \beta_1, \beta_2} \frac{I + II + III + IV}{E_{\alpha_1} - E_{\beta_1} + E_{\alpha_2} - E_{\beta_2}}.$$

Using the fact that operators acting on different ions commute, we can simplify the four sums and split them into a sum over pairs of ions and a sum over triplets of ions as follows,

$$\frac{I}{E_D^2} = 2 \sum_{i \neq j} V_{ij}^{\nu\mu} V_{ij}^{\sigma\tau} \left\{ P_i^{\alpha_1} J_i^\mu P_i^{\beta_1} J_i^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\tau \right] + \left[P_j^{\alpha_1} J_j^\nu P_j^{\beta_1}, J_j^\tau \right] J_i^\sigma P_i^{\alpha_2} J_i^\mu P_i^{\beta_2} \right\} \\ + 4 \sum_{i \neq j \neq k} V_{ij}^{\nu\mu} V_{kj}^{\sigma\tau} P_i^{\alpha_1} J_i^\mu P_i^{\beta_1} J_k^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\tau \right] \\ \frac{II}{E_D J_{ex}} = \frac{1}{12} \sum_{i \neq j} V_{ij}^{\nu\mu} \delta_{ij, nn} \left\{ P_i^{\alpha_1} J_i^\mu P_i^{\beta_1} J_i^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\sigma \right] + J_i^\sigma P_i^{\alpha_2} J_i^\nu P_i^{\beta_2} \left[P_j^{\alpha_1} J_j^\mu P_j^{\beta_1}, J_j^\sigma \right] \right\} \\ + \frac{1}{6} \sum_{i \neq j \neq k} V_{ij}^{\nu\mu} \delta_{kj, nn} P_i^{\alpha_1} J_i^\mu P_i^{\beta_1} J_k^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\sigma \right]$$

$$\begin{aligned}
\frac{III}{E_D J_{\text{ex}}} &= \frac{1}{12} \sum_{i \neq j} V_{ij}^{\sigma\tau} \delta_{ij,nn} \left\{ P_i^{\alpha_1} J_i^\nu P_i^{\beta_1} J_i^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\tau \right] + \left[P_j^{\alpha_1} J_j^\nu P_j^{\beta_1}, J_j^\sigma \right] J_i^\tau P_i^{\alpha_2} J_i^\nu P_i^{\beta_2} \right\} \\
&\quad + \frac{1}{6} \sum_{i \neq j \neq k} V_{kj}^{\sigma\tau} \delta_{ij,nn} P_i^{\alpha_1} J_i^\nu P_i^{\beta_1} J_k^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\tau \right] \\
\frac{IV}{J_{\text{ex}}^2} &= \frac{1}{36} \sum_{i \neq j} \delta_{ij,nn} \left\{ P_i^{\alpha_1} J_i^\nu P_i^{\beta_1} J_i^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\sigma \right] + \left[P_j^{\alpha_1} J_j^\nu P_j^{\beta_1}, J_j^\sigma \right] J_i^\sigma P_i^{\alpha_2} J_i^\nu P_i^{\beta_2} \right\} \\
&\quad + \frac{1}{18} \sum_{i \neq j \neq k} \delta_{ij,nn} \delta_{kj,nn} P_i^{\alpha_1} J_i^\nu P_i^{\beta_1} J_k^\sigma \left[P_j^{\alpha_2} J_j^\nu P_j^{\beta_2}, J_j^\sigma \right]
\end{aligned}$$

Since S is block-off-diagonal by construction, its commutator with the block-diagonal part of H_T necessarily lies outside of the low-energy block and would be eliminated by the projection operators, that is, $\frac{1}{2}P_0[S, (H_T)_d]P_0 = 0$. Therefore $\frac{1}{2}P_0[S, (H_T)_{od}]P_0 = \frac{1}{2}P_0[S, H_T]P_0$, and the latter is used in the main text.

The sums over operators (μ, ν, τ, σ) and crystal field levels ($\alpha_1, \alpha_2, \beta_1, \beta_2$) are calculated for a given B_x using *Mathematica*.

For the three-state model in zero-field, the above commutators result in $H_{\text{eff}}(B_x = 0)$, as given in the main text. It includes effective fields h_i^μ and two-body interaction $\varepsilon_{ij}^{\mu\nu}$, which are given here in terms of the parameters Δ, ρ , and α , described in the main text. Elements not specified are zero.

$$\begin{aligned}
\varepsilon_{ij}^{xx} &= \frac{\rho^4 E_D^2}{4\Delta} \left(2(V_{ij}^{xy})^2 - (V_{ij}^{xx})^2 - (V_{ij}^{yy})^2 \right) \\
&\quad - \frac{\rho^4}{2\Delta} J_{\text{ex}} \delta_{ij,nn} [E_D (V_{ij}^{xx} + V_{ij}^{yy}) + J_{\text{ex}}]
\end{aligned} \tag{S.5}$$

$$\begin{aligned}
\varepsilon_{ij}^{yy} &= -\frac{\rho^4 E_D^2}{2\Delta} \left((V_{ij}^{xy})^2 + V_{ij}^{xx} V_{ij}^{yy} \right) \\
&\quad - \frac{\rho^4}{2\Delta} J_{\text{ex}} \delta_{ij,nn} [E_D (V_{ij}^{xx} + V_{ij}^{yy}) + J_{\text{ex}}]
\end{aligned} \tag{S.6}$$

$$\varepsilon_{ij}^{zz} = \frac{\rho^4 E_D^2}{2\Delta} \left(V_{ij}^{xx} V_{ij}^{yy} - (V_{ij}^{xy})^2 \right) \tag{S.7}$$

$$\begin{aligned}
&\quad + \frac{\rho^4}{2\Delta} J_{\text{ex}} \delta_{ij,nn} [E_D (V_{ij}^{xx} + V_{ij}^{yy}) + J_{\text{ex}}]
\end{aligned} \tag{S.8}$$

$$\begin{aligned}
\varepsilon_{ij}^{xy} &= \frac{\rho^4 E_D^2}{2\Delta} (V_{ij}^{yy} V_{ij}^{xy} - V_{ij}^{xx} V_{ij}^{xy}) \\
\varepsilon_{ij}^{xy} &= \varepsilon_{ij}^{yx}
\end{aligned}$$

$$\begin{aligned}
h_i^x &= \frac{\alpha^2 \rho^2 E_D^2}{\Delta} \sum_{j(\neq i)} \left((V_{ij}^{yz})^2 - (V_{ij}^{xz})^2 \right) \\
&\quad + \frac{\rho^4 E_D^2}{2\Delta} \sum_{j(\neq i)} \left((V_{ij}^{yy})^2 - (V_{ij}^{xx})^2 \right) \\
&\quad + \frac{J_{\text{ex}} \rho^4 E_D}{\Delta} \delta_{ij,nn} (V_{ij}^{yy} - V_{ij}^{xx})
\end{aligned} \tag{S.9}$$

$$\begin{aligned}
h_i^y &= -\frac{2\alpha^2 \rho^2}{\Delta} E_D^2 \sum_{j(\neq i)} (V_{ij}^{xz} V_{ij}^{yz}) \\
&\quad - \frac{\rho^4}{\Delta} E_D^2 \sum_{j(\neq i)} V_{ij}^{xy} (V_{ij}^{xx} + V_{ij}^{yy}) \\
&\quad - \frac{2J_{\text{ex}} \rho^4}{\Delta} E_D \delta_{ij,nn} V_{ij}^{xy}
\end{aligned} \tag{S.10}$$

$$h_i^z = 0 \quad (\text{S.11})$$

These expressions coincide¹ with those given in the appendix to Ref. [31] when J_{ex} is taken to be zero.

II. MEAN-FIELD APPROXIMATION AT $B_x = 0$

We employ a mean-field (MF) approximation by neglecting correlation terms $(\sigma_i^\mu - \langle \sigma_i^\mu \rangle)(\sigma_j^\nu - \langle \sigma_j^\nu \rangle) \approx 0$ for any μ, ν and $i \neq j$, and denoting $\langle \sigma_j^\mu \rangle \equiv m_\mu$. The resulting single-body mean-field Hamiltonian is given by

$$H_{\text{MF}} = - \sum_i b_i^\mu \sigma_i^\mu \quad (\text{S.12})$$

Where the local fields b_i^μ , simplified by employing the S_4 symmetry of the crystal about the Ho^{3+} ions and denoting the number of nearest neighbors q , are given by

$$\begin{aligned} b_i^x = & \frac{\rho^4 J_{\text{ex}}}{\Delta} m_x \left(2E_D \sum_{j(\neq i)} \delta_{ij,nn} V_{ij}^{xx} + qJ_{\text{ex}} \right) \\ & + \frac{\rho^4}{\Delta} E_D^2 m_x \sum_{j(\neq i)} \left[(V_{ij}^{xx})^2 - (V_{ij}^{xy})^2 \right] \\ & - \frac{2\rho^4}{\Delta} E_D^2 m_y \sum_{j(\neq i)} V_{ij}^{yy} V_{ij}^{xy} \end{aligned} \quad (\text{S.13})$$

$$\begin{aligned} b_i^y = & \frac{\rho^4 J_{\text{ex}}}{\Delta} m_y \left(2E_D \sum_{j(\neq i)} \delta_{ij,nn} V_{ij}^{xx} + qJ_{\text{ex}} \right) \\ & + \frac{\rho^4}{\Delta} E_D^2 m_y \sum_{j(\neq i)} \left[(V_{ij}^{xy})^2 + V_{ij}^{xx} V_{ij}^{yy} \right] \\ & - \frac{2\rho^4}{\Delta} E_D^2 m_x \sum_{j(\neq i)} V_{ij}^{yy} V_{ij}^{xy} \end{aligned} \quad (\text{S.14})$$

$$\begin{aligned} b_i^z = & -\alpha^2 m_z E_D \sum_{j(\neq i)} V_{ij}^{zz} - \alpha^2 q J_{\text{ex}} m_z \\ & - \frac{\rho^4 J_{\text{ex}} m_z}{\Delta} \left(2E_D \sum_{j(\neq i)} \delta_{ij,nn} V_{ij}^{xx} + qJ_{\text{ex}} \right) \\ & - \frac{\rho^4 m_z}{\Delta} E_D^2 \sum_{j(\neq i)} \left[V_{ij}^{xx} V_{ij}^{yy} - (V_{ij}^{xy})^2 \right] \\ & + \frac{4\alpha^2 \rho^2 m_z}{\Delta} E_D^2 \sum_{k \neq j(\neq i)} V_{ij}^{xz} V_{jk}^{xz} \end{aligned} \quad (\text{S.15})$$

The self-consistency equations for a spin- $\frac{1}{2}$ in an arbitrary mean-field $\vec{b}(\vec{m})$ at temperature $k_B T = \beta^{-1}$ are [36]

$$m_\mu = \frac{b^\mu}{|\vec{b}|} \tanh \left(\beta |\vec{b}| \right)$$

¹ Except for a factor of half between the second term in Eq. (S.10) here and Eq. (23) in Ref. [31], likely due to a misprint. This term

vanishes for the pure system, so the difference is inconsequential.

and it is readily observed that $m_x = m_y = 0$ satisfies the first two equations, leaving us with the usual Ising self-consistency equation

$$m_z = \tanh(\beta b_i^z)$$

from which we get T_c given by Eq. (7) in the main text.

Lattice Sums

In order to get a numerical estimation of T_c , we need to calculate the sums in Eq. (7) in the main text and similar sums arising at $B_x > 0$. We define

$$A_{\mu\nu} = \sum_{j(\neq i)} V_{ij}^{\mu\nu}$$

$$B_{\mu\nu,\sigma\tau} = \sum_{j(\neq i)} V_{ij}^{\mu\nu} V_{ij}^{\sigma\tau}$$

and express the various sums that appear in the mean fields derived from either the three-state or 17-state models using these definitions,

$$\begin{aligned} \sum_{j \neq k(\neq i)} V_{ki}^{\mu\nu} V_{kj}^{\sigma\tau} &= \sum_{k(\neq i)} V_{ki}^{\mu\nu} \left(\sum_{j(\neq k \neq i)} V_{kj}^{\sigma\tau} \right) \\ &= \sum_{k(\neq i)} V_{ki}^{\mu\nu} \left(-V_{ki}^{\sigma\tau} + \sum_{j(\neq k)} V_{kj}^{\sigma\tau} \right) \\ &= \sum_{k(\neq i)} V_{ki}^{\mu\nu} (-V_{ki}^{\sigma\tau} + A_{\sigma\tau}) \\ &= A_{\sigma\tau} A_{\mu\nu} - B_{\mu\nu,\sigma\tau}. \end{aligned} \tag{S.16}$$

Similarly, we also have $\sum_{j \neq k(\neq i)} V_{ij}^{\mu\nu} V_{ik}^{\sigma\tau} = A_{\sigma\tau} A_{\mu\nu} - B_{\mu\nu,\sigma\tau}$.

As the summand in $B_{\mu\nu,\sigma\tau}$ is $\sim 1/r^6$, it converges absolutely and rapidly, and is thus best numerically evaluated directly (values summarized in Supplementary Table I). The sum $A_{\mu\nu}$ is a bit more subtle; since it is $\sim 1/r^3$, it is conditionally convergent, meaning its value depends on the summation order—that is, it depends on the limiting sample shape.

Magnetic domains forming below T_c in a pattern of long thin cylinders (or needles) parallel to the easy axis [37, 38] are understood as the way the system minimizes stray fields due to their energetic cost [39]. This is the justification for the use of an infinitely long thin cylinder as the limiting shape in the mean-field approximation when calculating A_{zz} [28, 29, 40, 41]. Nonetheless, as noted by Chakraborty *et al.* [28], the transverse component of the magnetization, which develops in response to the applied transverse field, is indifferent to this domain structure. Therefore, the shape-dependent transverse dipolar interactions, A_{xx} and A_{yy} , should be calculated by summing over the entire sample and not just a single needle-shaped domain as with A_{zz} . Assuming the principal axes of the sample are aligned with the single-ion magnetic axes, the off-diagonal sums vanish, so $A_{\mu\nu} = 0$ for $\mu \neq \nu$.

To calculate these sums $A_{\mu\mu}$, we use Ewald's method to obtain their Fourier transform and evaluate it at $\vec{k} \rightarrow 0$. The Fourier transform of the dipolar interaction for a lattice with a basis is given by [49]

$$\begin{aligned} V_{\vec{k}}^{\alpha\beta} &= \sum_{\vec{\tau}} \frac{4\pi}{v} \frac{k^\alpha k^\beta}{k^2} \exp(-k^2/(4R^2)) + \frac{4\pi}{v} \sum_{\vec{\tau}} \sum_{\vec{G} \neq 0} \frac{(G^\alpha + k^\alpha)(G^\beta + k^\beta)}{|\vec{G} + \vec{k}|^2} \exp\left(-|\vec{G} + \vec{k}|^2/(4R^2) - i\vec{G} \cdot \vec{\tau}\right) \\ &\quad - \sum_{\vec{\tau}} \sum_{\vec{r}_l} \left\{ \frac{2R \exp(-R^2(\vec{r}_l + \vec{\tau})^2)}{\sqrt{\pi}(\vec{r}_l + \vec{\tau})^2} \left[\left(3 + 2R^2(\vec{r}_l + \vec{\tau})^2\right) \frac{(r_l^\alpha + \tau^\alpha)(r_l^\beta + \tau^\beta)}{(\vec{r}_l + \vec{\tau})^2} - \delta_{\alpha\beta} \right] \right. \\ &\quad \left. + \left(\frac{3(r_l^\alpha + \tau^\alpha)(r_l^\beta + \tau^\beta)}{(\vec{r}_l + \vec{\tau})^5} - \frac{\delta_{\alpha\beta}}{(\vec{r}_l + \vec{\tau})^3} \right) \operatorname{erfc}(R|\vec{r}_l + \vec{\tau}|) \right\} \exp(-\vec{k} \cdot \vec{r}_l) - \frac{4R^3}{3\sqrt{\pi}} \delta_{\alpha\beta} \end{aligned} \tag{S.17}$$

where $\vec{\tau}$ are the vectors indicating the locations of atoms in the basis, $\vec{G}$ are vectors of the reciprocal lattice, and $\vec{r}_l$ are vectors of the real underlying Bravais lattice. v is the volume of a unit cell, and R is a parameter used to balance the convergence of the real and reciprocal sums, usually set to $R = \frac{2}{a}$. The prime on the sum indicates that the origin is excluded, i.e., $\vec{\tau} \neq 0$ when $\vec{r}_l = 0$. It is important to note that the first term in (S.17) is non-analytic as $\vec{k} \rightarrow 0$ —a manifestation of the shape dependence discussed above. From a macroscopic point of view, the shape dependence of the sums $A_{\mu\nu}$ translates to a shape-dependent demagnetization factor. Defining the demagnetization factors as [42]

$$L_\mu = 4\pi \lim_{\vec{k} \rightarrow 0} \left(\frac{k_\mu}{k} \right)^2$$

and using the longitudinal factor $L_z = 0$ (appropriate to a needle-shaped domain) and the transverse factors $L_x = L_y = \frac{4\pi}{3}$ (appropriate to a spherical sample) [39], we obtain

$$A_{zz} = V_{\vec{k}=0}^{zz} = -11.271 [\text{\AA}^{-3}] \quad (\text{S.18})$$

$$A_{xx} = A_{yy} = V_{\vec{k}=0}^{xx} = 1.603 [\text{\AA}^{-3}] \quad (\text{S.19})$$

where distances are measured in units of $a = 5.175 \text{\AA}$. These are consistent with the values in Ref. [33], subtracting the respective demagnetization factors.

Supplementary Table I. Values of $B_{\mu\nu, \sigma\tau} = \sum_{j(\neq i)} V_{ij}^{\mu\nu} V_{ij}^{\sigma\tau}$ calculated by direct summation; values not specified are zero. $a = 5.175 \text{\AA}$ is the length of the LiHoF₄ unit cell.

$B_{zz,zz}$	$17.93 \text{\AA}^{-6}$
$B_{xx,yy}, B_{yy,xx}$	$-22.05 \text{\AA}^{-6}$
$B_{yy,yy}, B_{xx,xx}$	$31.02 \text{\AA}^{-6}$
$B_{xz,xz}, B_{xz,zx}, B_{zx,xz}, B_{zx,zx},$ $B_{yz,yz}, B_{yz,zy}, B_{zy,yz}, B_{zy,zy}$	$36.73 \text{\AA}^{-6}$
$B_{xy,xy}, B_{xy,yx}, B_{yx,xy}, B_{yx,yx}$	$4.80 \text{\AA}^{-6}$
$B_{yy,zz}, B_{zz,yy}, B_{xx,zz}, B_{zz,xx}$	$-8.96 \text{\AA}^{-6}$

Sums of the form $\sum_{j(\neq i)} V_{ij}^{\mu\nu} \delta_{ij,nn}$ are trivially calculated by summing the four dipolar terms associated with the four nearest neighbors.

In the case of $B_x > 0$, we also encounter slightly more subtle sums of the form $\sum_{j \neq k(\neq i)} V_{ij}^{\mu\nu} \delta_{jk,nn}$, which can be expressed as

$$\sum_{j \neq k(\neq i)} V_{ij}^{\mu\nu} \delta_{jk,nn} = q \sum_{j(\neq i)} V_{ij}^{\mu\nu} - \sum_{j(\neq i)} \delta_{ij,nn} V_{ij}^{\mu\nu}$$

Eventually, combining Eq. (S.16), the values listed in Supplementary Table I, and the values given in Eqs. (S.18) and (S.19), we can evaluate Eq. (7) from the main text to find $T_c = 2.18 \text{ K}$, which is about 5% lower than the critical temperature found from applying a mean-field approximation directly to the microscopic Hamiltonian ($T_c = 2.27 \text{ K}$).

An interesting consequence of Eq. (S.16) is that the three-body sum in Eq. (7) in the main text, $\sum_{k \neq j(\neq i)} V_{ij}^{xz} V_{jk}^{xz}$, is directly proportional to $B_{xz,xz}$. This is due to the mirror symmetry through a plane parallel to the $y-z$ plane which contains spin j . As illustrated in Supplementary Figure S1, this symmetry guarantees that a contribution to the field at spin i , by any spin k located at $\vec{r}_{ij} + (x_{jk}, y_{jk}, z_{jk})$ is offset by another spin k located at $\vec{r}_{ij} + (-x_{jk}, y_{jk}, z_{jk})$. The only contribution not reduced by this symmetry comes from the spin k that is located at $\vec{r}_{ij} + (x_{ij}, -y_{ij}, -z_{ijk}) = (2x_{ij}, 0, 0)$, since its counterpart is omitted from the sum by the restriction $i \neq k$. Consequently, a MF approximation only considers three-body interactions where the third spin is an equal distance, along the x -axis, from the two other spins. This explains why a MC simulation shows a greater reduction of T_c than seen in MF (see Fig. 2 in the main text); in MC not all spins are in the same state, so cancellations due to the mirror symmetry are not guaranteed.

III. MEAN-FIELD APPROXIMATION AT $B_x \geq 0$

As mentioned in the main text, the procedure described in Section I can be performed for arbitrary non-zero B_x . When a mean-field approximation is applied to the resultant effective Hamiltonian, the three resulting self-consistency

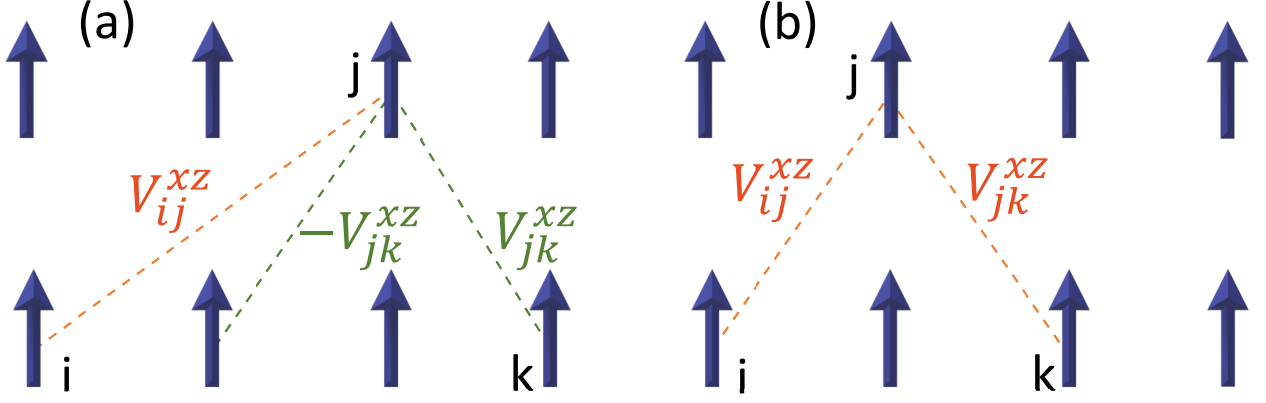


FIG. S1. An illustration demonstrating the cancellations occurring in mean-field. (a) For a given pair of spins i and j , the three-body interaction involving a typical third spin k , $V_{ij}^{xz}V_{jk}^{xz}$, would tend to offset by the three-body interaction involving the mirror image of k relative to j (indicated by $-V_{jk}^{xz}$). (b) The exception where the interactions do not cancel is when k is chosen such that its mirror image is V_{ij}^{xz} , which is omitted from the sum by the restriction $k \neq i$. The cancellations are only guaranteed in MF, where the system is assumed homogeneous, but not in a MC simulation or, indeed, the physical system.

equations are fully coupled. Thus, to find T_c , self-consistency equations are solved numerically for decreasing T until a non-zero value for $\langle J^z \rangle$ is detected. At the low-temperature regime, where the critical field is nearly independent of temperature (when the hyperfine interaction is not considered), we instead calculate $\langle J^z \rangle$ for various B_x at a given temperature and fit $\langle J^z \rangle \propto \sqrt{B_x^c - B_x}$ to find the critical field B_x^c . We note that in this new basis of $|\alpha\rangle$ and $|\beta\rangle$, we are no longer interested simply in $\langle \sigma_i^z \rangle \equiv m_z$ since it no longer represents magnetization along the physical z -axis. Instead, for each value B_x , we decompose the two-dimensional projection of J^μ , as done in Ref. [28],

$$J^\mu = C_\mu + \sum_{\nu=x,y,z} C_{\mu\nu}(B_x) \sigma^\nu. \quad (\text{S.20})$$

Then, after self-consistently obtaining m_x , m_y , and m_z , we can plot $\langle J^\mu \rangle = C_{\mu z}m_z + C_{\mu y}m_y + C_{\mu x}m_x + C_\mu$ vs. temperature. These values are plotted for $B_x = 1$ T in Supplementary Figure S2.

It might seem initially surprising that $\langle J_x \rangle$ is almost independent of T , also at $T > T_c$, as seen in Supplementary Figure S2(b). This behavior is expected in the ferromagnetic phase of the transverse-field Ising model, but as $T \rightarrow \infty$, all expectation values should tend towards zero (as seen in Supplementary Figure S2(a)). This apparent issue is resolved by recognizing that the decomposition (S.20) is only valid at low temperatures.

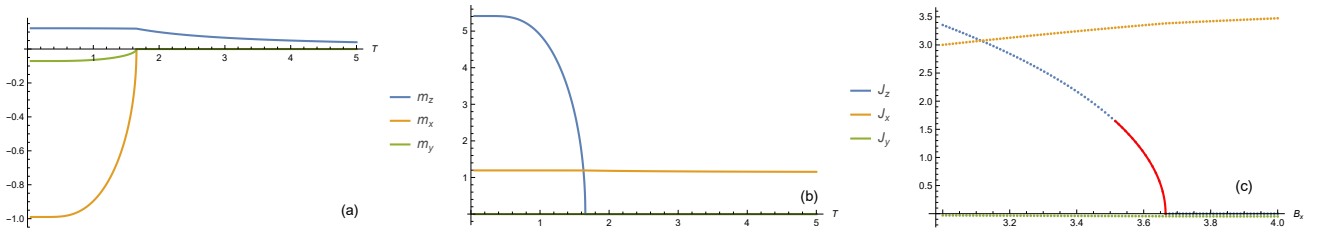


FIG. S2. Magnetization vs. Temperature and Transverse field. (a) Raw solutions of the self-consistency equations, m_x , m_y , and m_z , as a function of temperature at $B_x = 1$ T; (b) composite physical values, J_x , J_y , and J_z , found from the raw solutions by (S.20). (c) Physical expectation values as a function of B_x at $T = 0$. The solid red line is a best-fit of $\langle J^z \rangle \propto \sqrt{B_x^c - B_x}$. Due to the relative numerical complexity, the latter approach is used only at $T \leq 0.3$ K.

IV. MONTE CARLO SIMULATION

The Hamiltonian used in MC simulations is very similar to $H_{\text{eff}}(B_x = 0)$ given by Eq. (5) in the main text, with two exceptions. The first, described in the main text, is that quantum terms, i.e., terms involving σ^x or σ^y operators, are

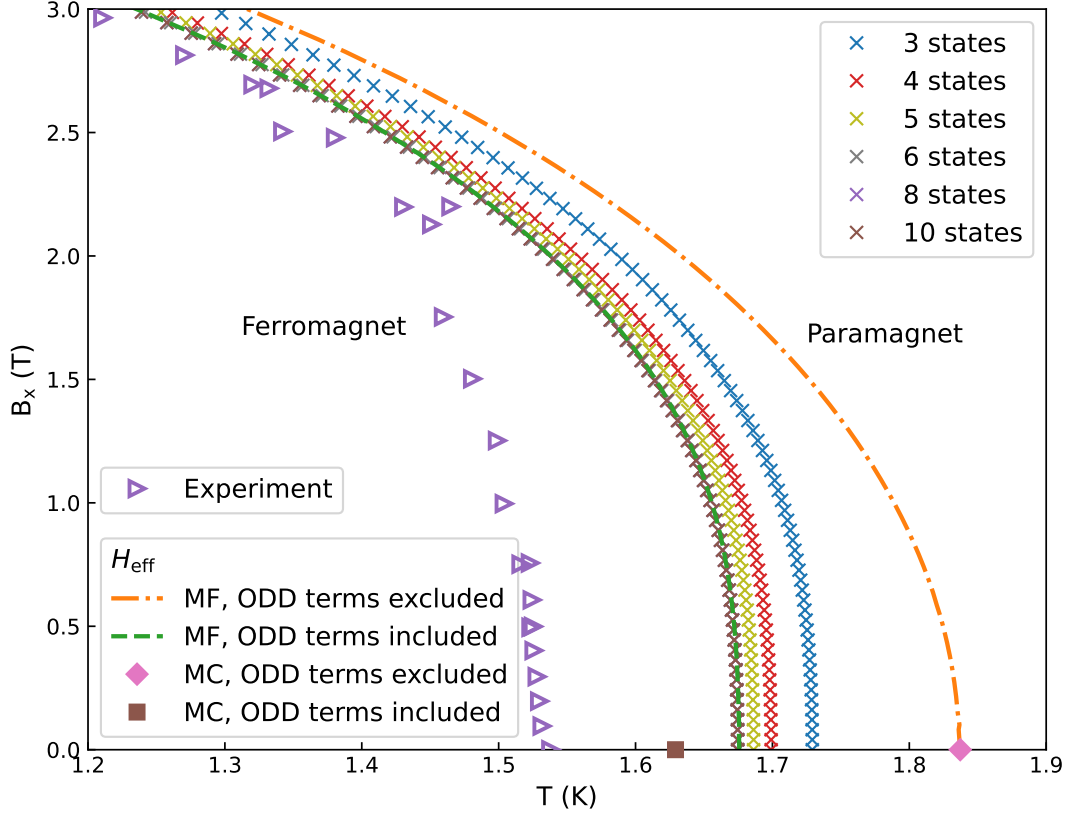


FIG. S3. A subset of the $B_x - T$ phase diagram of LiHoF_4 illustrating the effect of including more excited states in the derivation of the effective Hamiltonian. X's of different colors represent MF results of H_{eff} with a different number of states considered. The three-state model, described in detail in the main text, shows the smallest reduction, while, evidently, a 6-state model already sufficiently captures the full effect shown by the 17-state model used in this work (indicated by a dashed green line).

omitted. The second is that the effective Hamiltonian is derived from the 17-state model rather than the three-state model. Since the quantum terms have a vanishing MF contribution, the Hamiltonian used in MC simulations is essentially the same as the one used to obtain the MF results shown in Fig. 2 in the main text. The two differences introduced to the classical terms by switching to the 17-state model are in the magnitudes of the coefficients and in the emergence of a secondary, diagonal, three-body interaction of magnitude Γ_D , so

$$\begin{aligned}
 H_{\text{MC}} = & \frac{\alpha^2}{2} E_D \sum_{i \neq j} V_{ij}^{zz} \sigma_i^z \sigma_j^z + \frac{\alpha^2}{2} J_{\text{ex}} \sum_{i \neq j} \sigma_i^z \sigma_j^z \\
 & - \Gamma E_D^2 \sum_{i \neq j \neq k} \left(V_{ik}^{xz} V_{jk}^{xz} + V_{ik}^{yz} V_{jk}^{yz} \right) \sigma_i^z \sigma_j^z \\
 & - \Gamma_D E_D^2 \sum_{i \neq j \neq k} V_{ik}^{zz} V_{jk}^{zz} \sigma_i^z \sigma_j^z \\
 & + \sum_{i \neq j} \tilde{\epsilon}_{ij}^{zz} \sigma_i^z \sigma_j^z
 \end{aligned} \tag{S.21}$$

where

$$\Gamma = \alpha^2 \sum_{i=3}^{17} \frac{\langle i | J^x | \uparrow \rangle \langle \uparrow | J^x | i \rangle}{\langle i | V_c | i \rangle - \langle \uparrow | V_c | \uparrow \rangle} \approx 1.5 \frac{\alpha^2 \rho^2}{\Delta}$$

$$\tilde{\epsilon}_{ij}^{zz} = 1.623 \frac{\epsilon_{ij}^{zz}}{\rho^4/2\Delta} \approx 1.25 \epsilon_{ij}^{zz}$$

$$\Gamma_D \approx 0.27 \text{ K}^{-1}$$

Here we see that the inclusion of all 17 crystal-field states increases the magnitudes of the emergent three-body and two-body interactions by about 50% and 25%, respectively, compared with the inclusion of just one excited state. The cumulative effect of including higher excited states is demonstrated in Supplementary Figure S3. The secondary three-body interaction, of magnitude Γ_D , is almost negligible compared to the other three-body terms.

Three-Body Term in Periodic Boundary Conditions

The usage of the Ewald summation method in MC simulations where periodic boundary conditions are implemented is well known at this point, and we shall not fully recount it here. Instead, we provide only a brief summary of the details needed to justify how we incorporate the three-body dipolar (and hence long-range) term in (S.21) in a finite system with periodic boundary conditions.

Contrary to its usage in Section II, where we sum over all other spins in an *infinite* system, here Ewald's method is used to find an effective interaction between each pair of spins within a simulation box by summing over all periodic copies of said spins. Thus, for a finite system of size $L_x \times L_y \times L_z$ with periodic boundary conditions containing N spins, we may write

$$\begin{aligned} H_{\text{PBC}} &= \frac{1}{2} E_D \sum_{\nu, \mu} \sum_{i \neq j} \left[\frac{\delta^{\mu\nu} r_{ij}^2 - 3 r_{ij}^\mu r_{ij}^\nu}{r_{ij}^5} \right] \sigma_i^\mu \sigma_j^\nu \\ &= \frac{1}{2} E_D \sum_{\nu, \mu} \sum_{i=1}^N \sum_{j=1}^N \sum_{\vec{n}}' \left[\frac{\delta^{\mu\nu} |\vec{r}_{ij} + \vec{n}|^2}{|\vec{r}_{ij} + \vec{n}|^5} - \frac{3 (\vec{r}_{ij} + \vec{n})^\mu (\vec{r}_{ij} + \vec{n})^\nu}{|\vec{r}_{ij} + \vec{n}|^5} \right] \sigma_i^\mu \sigma_j^\nu \\ &\equiv \frac{1}{2} E_D \sum_{\nu, \mu} \sum_{i=1}^N \sum_{j=1}^N J_{ij}^{\mu\nu} \sigma_i^\mu \sigma_j^\nu \end{aligned}$$

where $\vec{r}_{ij} = \vec{r}_i - \vec{r}_j$, the sum over $\vec{n} = (n_x L_x, n_y L_y, n_z L_z)$ runs over all integers n_x, n_y, n_z . The prime indicates that the $\vec{n} = 0$ term is omitted for $i = j$. The challenge Ewald's method deals with is the calculation of $J_{ij}^{\mu\nu}$, which we shall assume resolved and refer the reader to Refs. [42, 43] and references therein for further information.

Hence, we are now concerned with finding the effective interaction between pairs of spins i and j that lie within the simulation box, which accounts for the interactions of these two spins and all their periodic copies. For concreteness, we consider the following three-body interaction term,

$$\begin{aligned} \sum_{k(\neq i, j)} V_{ik}^{xz} V_{jk}^{xz} &= \sum_{\vec{m}}' \sum_{\vec{n}}' \sum_{k=1}^N \frac{-3 (\vec{r}_{ik} + \vec{m})^x (\vec{r}_{ik} + \vec{m})^z}{|\vec{r}_{ik} + \vec{m}|^5} \times \frac{-3 (\vec{r}_{jk} + \vec{n})^x (\vec{r}_{jk} + \vec{n})^z}{|\vec{r}_{jk} + \vec{n}|^5} \\ &= \sum_{k=1}^N \left[\sum_{\vec{m}}' \frac{-3 (\vec{r}_{ik} + \vec{m})^x (\vec{r}_{ik} + \vec{m})^z}{|\vec{r}_{ik} + \vec{m}|^5} \right] \times \left[\sum_{\vec{n}}' \frac{-3 (\vec{r}_{jk} + \vec{n})^x (\vec{r}_{jk} + \vec{n})^z}{|\vec{r}_{jk} + \vec{n}|^5} \right] \\ &= \sum_{k=1}^N J_{ik}^{xz} J_{jk}^{xz}. \end{aligned}$$

Thus, given the effective interactions J_{ij}^{xz} calculated using Ewald's method, we obtain the effective interaction within the simulation box due to the three-body term by summing over all spins k within the simulation box, *including over i and j* (this accounts for cases where the third spin in an interaction is a copy of one of the other two spins).

Simulation Details

We use the parallel tempering Monte Carlo method [44]. To determine the critical temperature, we use the finite-size correlation length [45, 46],

$$\xi_L = \frac{1}{2 \sin(k_{\min}/2)} \left[\frac{\langle m^2(0) \rangle}{\langle m^2(\mathbf{k}_{\min}) \rangle} - 1 \right]^{\frac{1}{2}} \quad (\text{S.22})$$

where

$$m(\mathbf{k}) = \frac{1}{N} \sum_{i=1}^N \sigma_i^z \exp(-i\mathbf{k} \cdot \mathbf{R}_i). \quad (\text{S.23})$$

Here $\langle . \rangle$ refers to a thermal (MC) average. $\mathbf{R}_i$ is the location of the site i and $\mathbf{k}_{\min} = (\frac{2\pi}{L}, 0, 0)$. The finite-size correlation length divided by the linear system size L has a known scaling form,

$$\frac{\xi_L}{L} \sim \tilde{X} \left(L^{1/\nu} (T - T_c) \right) \quad (\text{S.24})$$

so that for $T = T_c$, it is independent of the system size L , and curves of different sizes should cross. Assuming the scaling function (S.24) is well approximated by a third-order polynomial close to the critical point, we perform a non-linear fit for the four polynomial coefficients, ν , and T_c . The crossing and fitting procedure are illustrated in Supplementary Figure S4(a) and Supplementary Figure S4(b), respectively. Statistical errors are estimated using the bootstrap method [47]. Finally, to extrapolate to infinite systems, we perform the above analysis for pairs of distinct system sizes $(L, 2L)$ and evaluate the crossing temperature $T_c(L, 2L)$ from each. The critical temperature in the thermodynamic limit is determined as the crossing of a linear fit to the $T_c(L, 2L)$ vs. $1/L$ data with the vertical axis, as shown in Supplementary Figure S4(d).

Equilibration of the simulation is verified by logarithmic binning of the data, i.e., the simulation time in terms of MC sweeps is successively increased by a factor of 2, and observables are averaged over that time. Once all observables of interest in three consecutive bins agree within error bars, the simulation is deemed equilibrated [47]. Supplementary Figure S4(c) shows the equilibration process of $\langle m^2(0) \rangle$. We run $N = 100$ independent simulations, each for 2^{12} MC sweeps for equilibration followed by another 2^{12} MC sweeps for measurement.

V. EFFECTIVE LOW-ENERGY DESCRIPTION OF THE Fe₈ MOLECULAR MAGNET

As described in the main text, the Fe₈ crystal consists of large molecular clusters, each described by a simple $S = 10$ spin model subject to a strong uniaxial anisotropy [3, 55, 59]. It is therefore well-described by the following Hamiltonian,

$$V_{\text{Fe}_8} = -DS_z^2 + E(S_x^2 - S_y^2) \quad (\text{S.25})$$

where $D/k_B = 0.294$ K and $E/k_B = 0.046$ K [55], defining the hard, medium, and easy axes. The structure of the Fe₈ crystal is triclinic with $a = 10.676$ Å, $b = 14.113$ Å, and $c = 15.147$ Å, and with $\alpha = 89.45^\circ$, $\beta = 109.96^\circ$, and $\gamma = 109.03^\circ$ [59]. The orientation of the magnetic anisotropy axes is determined in Ref. [59] and used in this work. It gives a magnetic easy-axis that creates an angle of around 15° with the crystallographic a -axis.

Projecting V_{Fe_8} and the angular momentum operators onto the four lowest-energy states of V_{Fe_8} , we get

$$\begin{aligned} V_{\text{Fe}_8} &= \begin{pmatrix} 0 & & & \\ & 0 & & \\ & & \Omega & \\ & & & \Omega \end{pmatrix}; & J^z &= \begin{pmatrix} -\beta & & & \\ & \beta & & \\ & & -\gamma & \\ & & & \gamma \end{pmatrix}; \\ J^x &= \begin{pmatrix} 0 & 0 & \sigma & 0 \\ 0 & 0 & 0 & -\sigma \\ \sigma & 0 & 0 & 0 \\ 0 & -\sigma & 0 & 0 \end{pmatrix}; & J^y &= \begin{pmatrix} 0 & 0 & -i\chi & 0 \\ 0 & 0 & 0 & -i\chi \\ i\chi & 0 & 0 & 0 \\ 0 & i\chi & 0 & 0 \end{pmatrix} \end{aligned} \quad (\text{S.26})$$

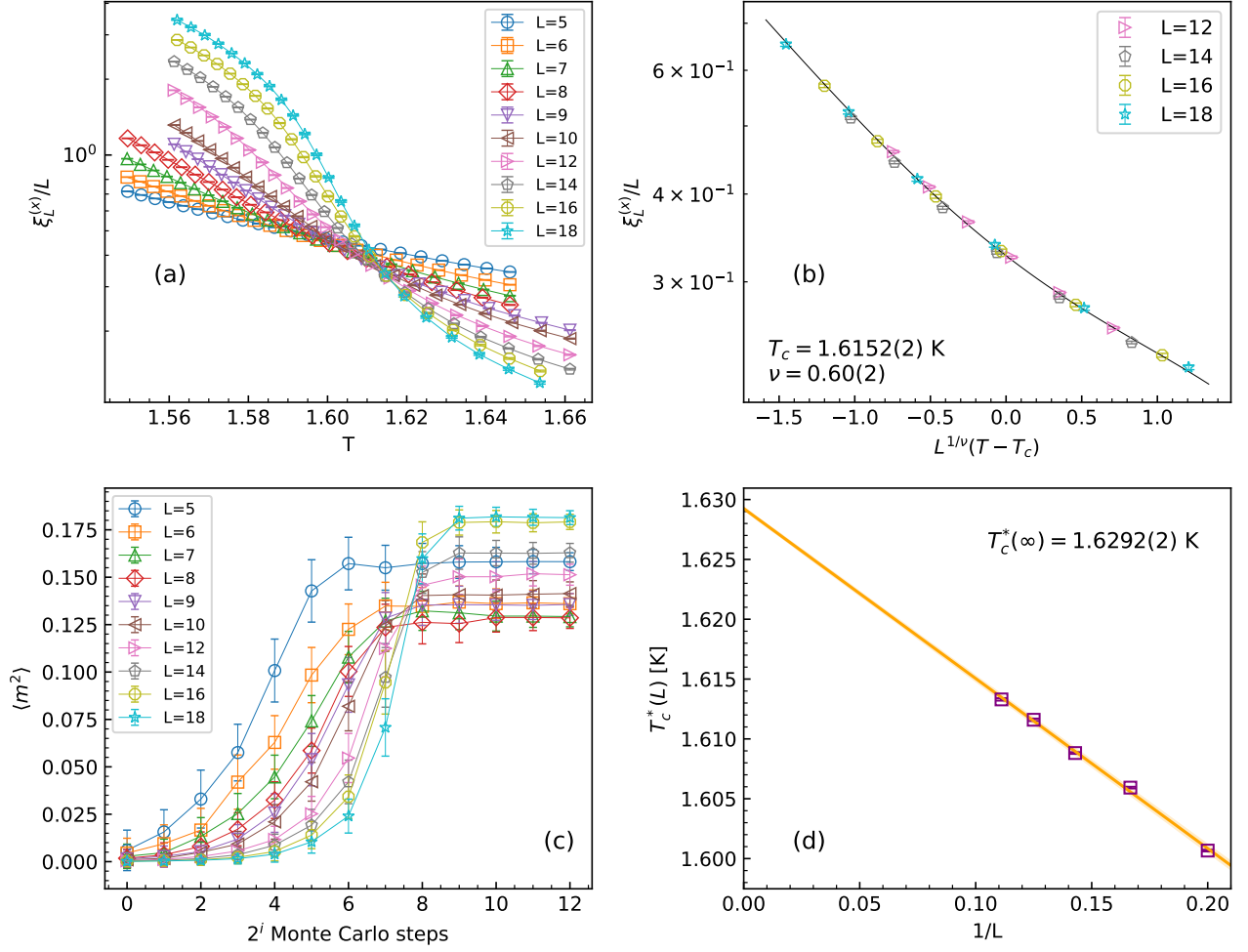


FIG. S4. Monte Carlo data for LiHoF₄ at $B_x = 0$. (a) Finite-size correlation length divided by linear system size L vs. temperature T . The clearly visible crossing indicates a phase transition. (b) Finite-size scaling analysis of the four largest system sizes, showing that, for an appropriate choice of parameters, data for different system sizes all fall on the same curve, indicated by the solid line (polynomial approximation). (c) Equilibration process of $\langle m^2(0) \rangle$ at $T \approx 1.58$ K. The i -th data point contains the average of 2^i MC sweeps, also averaged over independent simulation runs. (d) Extrapolation of T_c to the thermodynamic limit. Each data point represents an estimation of T_c obtained by finite-size scaling analysis of system sizes L and $2L$. A linear fit of the data gives $T_c(\infty) = 1.6295(2)$ K.

where the states are chosen such that $\sigma = 2.06$, $\chi = 2.43$, $\beta = 9.99$, and $\gamma = 8.97$ are real and $\Omega = 5.51$ K. The only non-negligible interaction between the Fe₈ molecular clusters is dipolar [58], so in the absence of an external transverse field

$$\mathcal{H}_{\text{Fe}_8} = \sum_i V_{\text{Fe}_8} + \frac{1}{2} E_{DF} \sum_{\substack{i \neq j \\ \nu, \mu}} V_{ij}^{\nu\mu} J_i^\nu J_j^\mu$$

with $E_{DF} = \frac{\mu_0 \mu_B^2 g^2}{4\pi}$ where $g = 2$ [55]. Applying the Schrieffer-Wolff transformation to obtain an effective low-energy spin- $\frac{1}{2}$ Hamiltonian, and then using a mean-field approximation, as before, gives a single self-consistency equation

that yields

$$\begin{aligned}
T_c &= -\beta^2 E_{DF} \sum_{j(\neq i)} V_{ij}^{zz} + \frac{2\beta^2 \sigma^2}{\Omega} E_{DF}^2 \sum_{j \neq k(\neq i)} V_{ij}^{xz} V_{jk}^{xz} \\
&\quad + \frac{2\beta^2 \chi^2}{\Omega} E_{DF}^2 \sum_{j \neq k(\neq i)} V_{ij}^{yz} V_{jk}^{yz} \\
&\quad + \frac{\sigma^2 \chi^2}{\Omega} E_{DF}^2 \sum_{j(\neq i)} \left((V_{ij}^{xy})^2 - V_{ij}^{xx} V_{ij}^{yy} \right) \\
&= -\beta^2 E_{DF} A_{zz} + \frac{2\beta^2 \sigma^2}{\Omega} E_{DF}^2 (A_{xz} A_{xz} - B_{xz, xz}) \\
&\quad + \frac{2\beta^2 \chi^2}{\Omega} E_{DF}^2 (A_{yz} A_{yz} - B_{yz, yz}) \\
&\quad + \frac{\sigma^2 \chi^2}{\Omega} E_{DF}^2 \sum_{j(\neq i)} (B_{xy, xy} - B_{xx, yy}) \\
&= 0.95 \text{ K}.
\end{aligned}$$

The $A_{\mu\nu}$'s and $B_{\mu\nu, \sigma\tau}$'s are calculated using Eq. (S.17). Indeed, the contribution of all but the first term above gives a negligible -0.06 mK , consistent with the lack of an observable discrepancy between MF and experiment.